

**FMUSER FUQ-2500B Analog RF Modulator 32-64 Channel
IP-HDMI-AV Input PAL-NTSC-SECAM Output
HEVC-H.265-H.264-MPEG-2 A2-NICAM728 OSD Insertion
Broadcast Systems**



❖ *Support LCD monitoring for output channels*

❖ *Support Channel/Volume control and management with knob and keyboard*



1U Chassis for 16in1&32in1



2U Chassis for 48in1&64in1



Product overview

The FMUSER FUQ-2500B converts IP/HDMI/AV inputs to analog RF outputs for up to 64 channels. It processes signals with full digital processing for high SFDR performance, supporting A2 and NICAM728 audio standards and PAL/NTSC/SECAM modulation including OSD insertion. Applications include television broadcasting for PAL, NTSC, or SECAM systems.

Key feature

- Supports A2 and NICAM728 audio standards with dual-channel mono and stereo modes.
- Ensures high SFDR performance through full digital processing for superior RF output.
- Provides OSD customization with logo/caption insertion, adjustable scrolling, and multilingual support per channel.
- Features hybrid input flexibility for 16-64 IP channels or HDMI/AV + IP combinations.
- Delivers multi-standard output with 32/64 carriers within 400MHz for PAL, NTSC, or SECAM broadcasts.

Support A2 and NICAM728 mono and stereo audio, up to dual channels

Audio options available:

Mono L, Mono R, Mono L+R

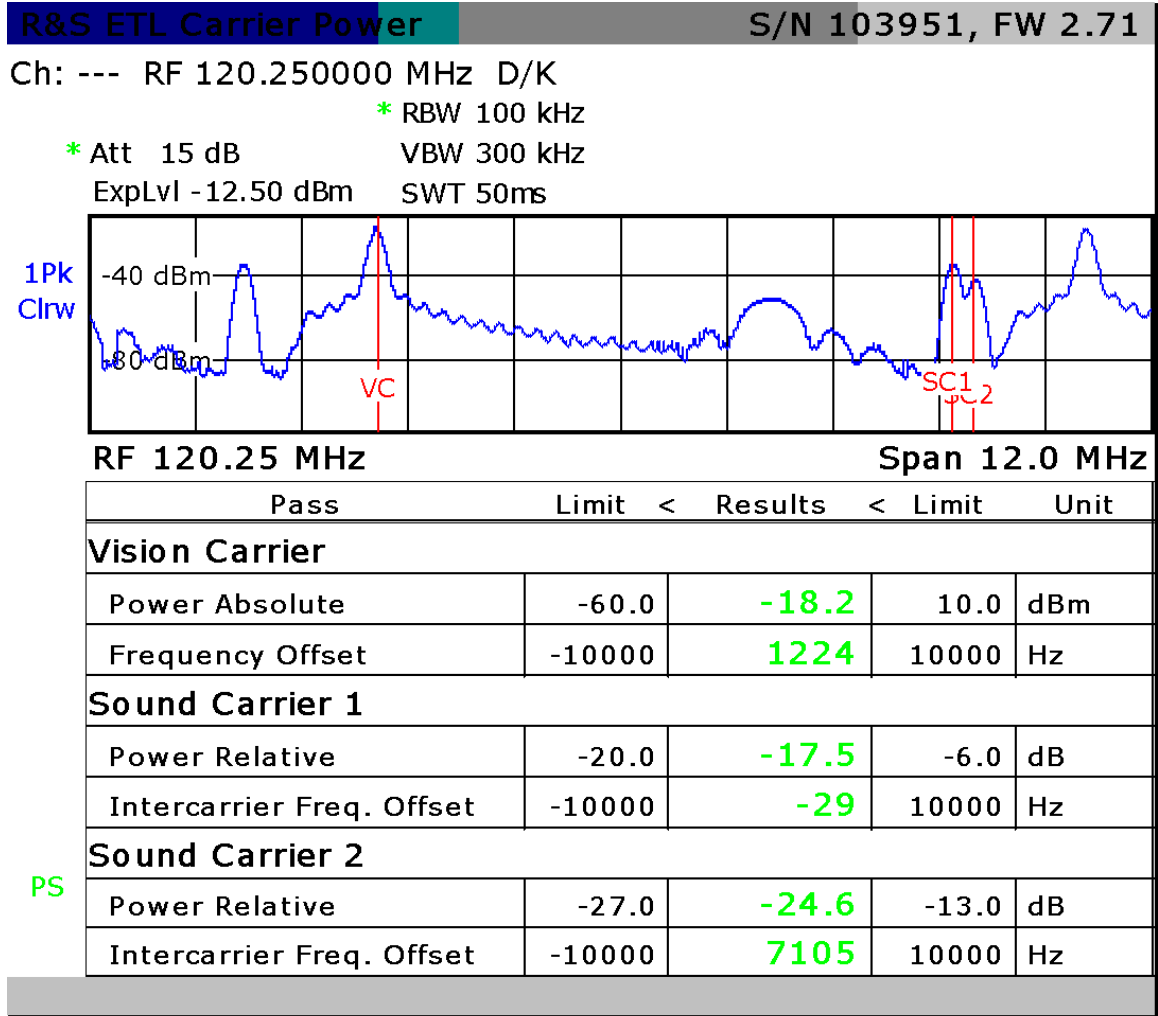
A2 Dual Mono (L, R), A2 Stereo

NICAM Dual Mono (L, R), NICAM Stereo

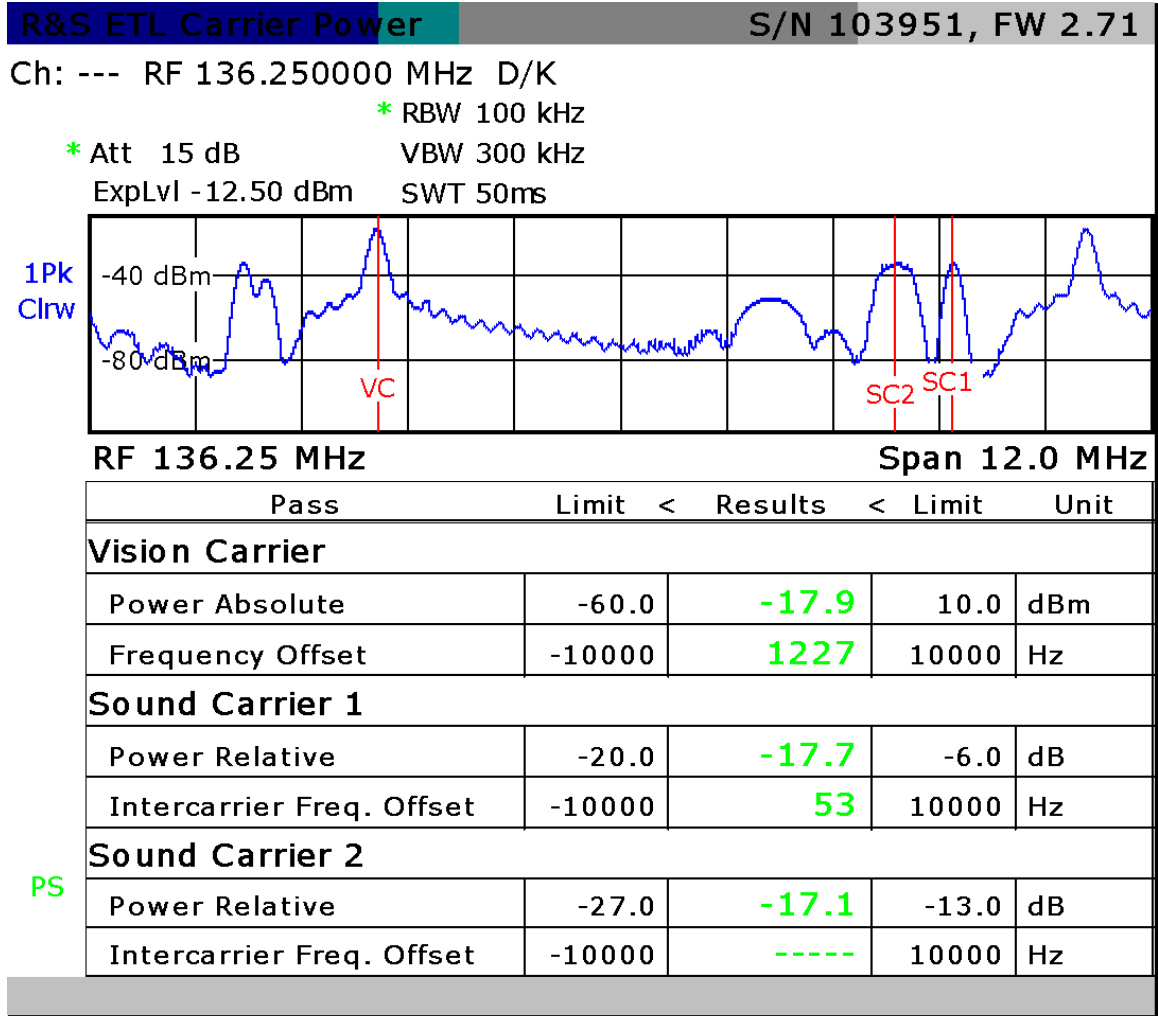
Knowledge about A2 and NICAM 728 audio:

NICAM 728: It is transmission of two-channel digital sound with terrestrial television systems B, G, H, I, K1 and L

A2 Stereo: Known as "two-channel sound", it is an analog television sound transmission system used in Germany, Austria, Australia, Switzerland, Netherlands and some other countries that use or used PAL-B or PAL-G. It relies on two separate FM carriers.



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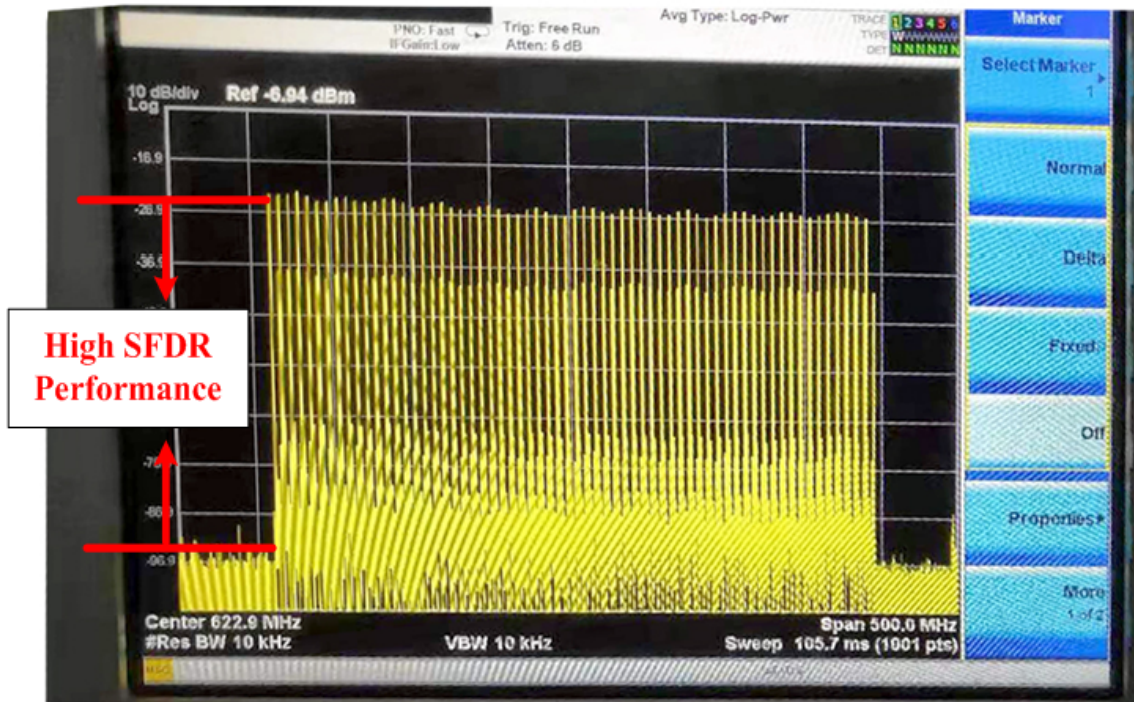


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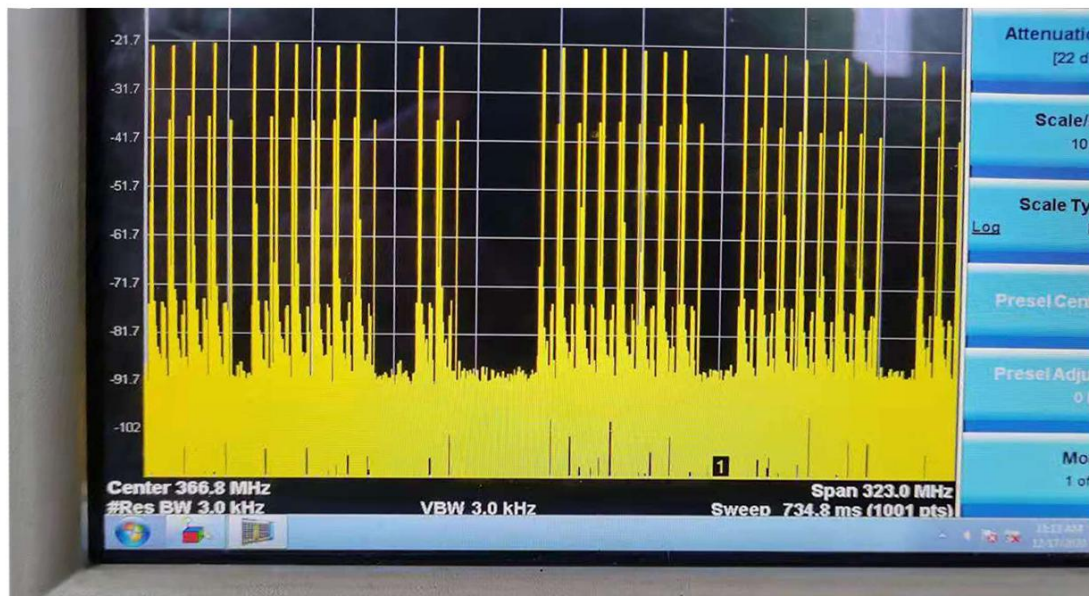
A2 Audio Spectral Diagram

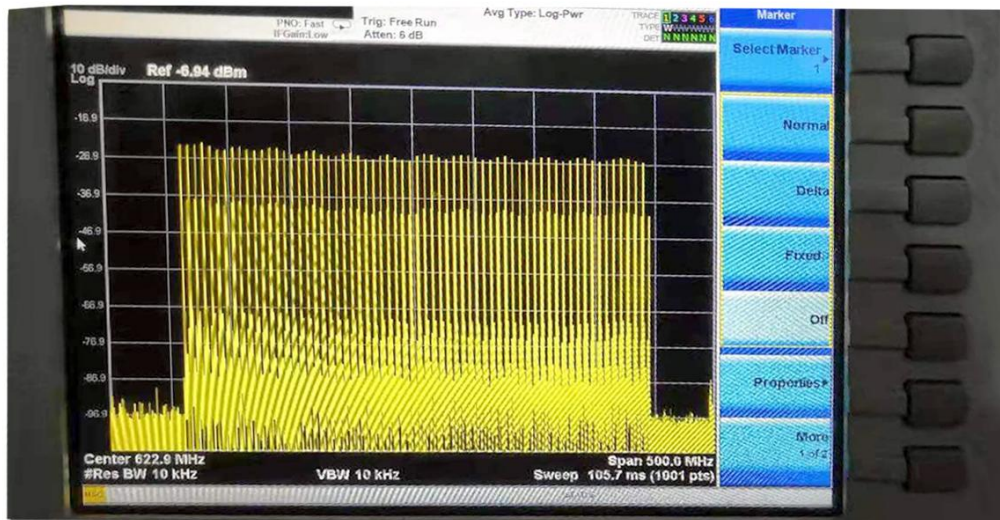
NICAM Audio Spectral Diagram

Support High SFDR Performance



Full Digital Processing, Super RF out performance





Support OSD (Logo&Caption) insertion for each channel, easy operation



Support Multiple-modulation Standard

32 (or 16/48/64) standard PAL or NTSC or SECAM TV programs output via 32 (or 16/48/64) non-adjacent or adjacent carriers output within 400MHz

Other Key feature

Support to convert 32 (or 16/48/64) IP channels (or 4 HDMI/AV+28 IP channels, or 8 HDMI/AV+24 IP channels) to analog RF out

Support CC/Subtitle/Teletext

Support BISS De-scrambling

Support program info modify, multiple audio select I

2 (or 4) GE ports for IP MPTS/SPTS input, Max 840Mbps for each GE input

Support HEVC/H.265, H.264/AVC, MPEG-2 TS Decoding

32 (or 16/48/64) standard PAL or NTSC or SECAM TV programs output via 32 (or 16/48/64) non-adjacent or adjacent carriers output within 400MHz

Redundancy power supply for option

Compact body, Freight saving

Support Web-based Network management

Specifications

Input	Interface/rate	32(or 16/48/64) IP (or 4 HDMI/AV+28 IP, or 8 HDMI/AV+24 IP) input 2 /4 GE ports (max 64 IP input) Max 840Mbps for each GE input
Input	Stream	UDP, UDP / RTP, 1-7 packets (SPTS/MPTS)
Input	Transport Protocol	UDP/RTP, unicast and multicast, IGMP V2/V3
Input	Packet Length	188 / 204 Bytes
Decoding Parameters	Video	HEVC/H.265, H.264/AVC, MPEG-2
Decoding Parameters	Audio	MPEG-1/2 Layer 1/2, (HE-)AAC
Decoding Parameters	Audio Channel	Mono L, Mono R, Mono L+R A2 Dual Mono (L, R), A2 Stereo NICAM Dual Mono (L, R), NICAM Stereo (Note: when applying dual audio, one TV program with dual audio will occupy 2 frequencies, which means the total output channels will reduce respectively.)
Decoding Parameters	Data	CC, Teletext, Subtitles, DVB Subtitling

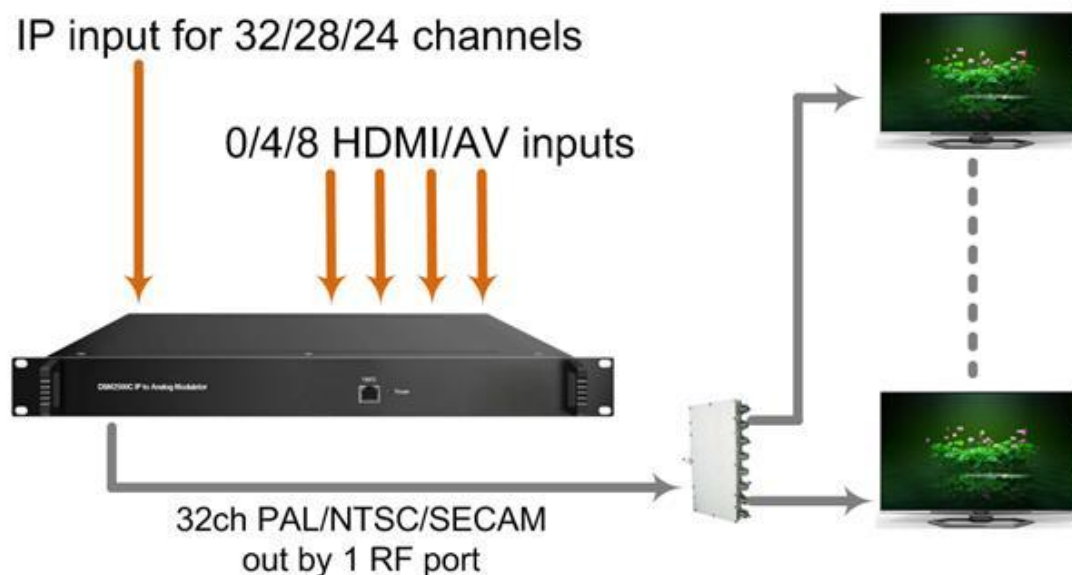
Decoding Parameters	Resolutions	HEVC/H.265: 1080@60P, 1080@60I, 1080@50P, 1080@50I, 720@60P, 720@50P H.264/AVC: 1080@60I, 1080@50P, 1080@50I, 1080@30P, 1080@25P, 720@60P, 720@50P, 576@50I, 480@60I MPEG2: 1080@60I, 1080@50I, 720@60P, 720@50P, 576@50I, 480@60I
Decoding Parameters	Aspect ratio	4:3
Modulation Parameters	Number of channels	32 (or 16/48/64) non-adjacent or adjacent carriers output
Modulation Parameters	Connectors	75Ω, F-jack
Modulation Parameters	Frequency range	43.25 – 951.252MHz
Modulation Parameters	Output Bandwidth	400MHz
Modulation Parameters	Output level	-5 ~ 2dBm (102 ~ 109dBμV)
Modulation Parameters	Return loss	≥ 14dB
Modulation Parameters	Spurious frequency dist.	≥ 60dB
Modulation Parameters	Residual carrier accuracy	1%
Modulation Parameters	TV standard	PAL B/G/D/K/M/N/I, NTSC M, SECAM
Modulation Parameters	Video-signal to noise ratio	≥ 60dB
Modulation Parameters	Carrier to noise ratio	≥ 60dB (when single carrier on) ≥ 55.5dB (when all 32 carriers on)
Network Interface	Management	1 x 100 Base-T Ethernet (RJ 45)
Network Interface	Data	2 x 1000 Base-T Ethernet (RJ 45)
Others	Image resolution output	480i/576i
Others	OSD	Logo : JPG, BMP, PNG
Others	OSD	Caption : support multiple languages, scrolling direction and speed customized
Others	Output volume adjustment	0 - 100 %

General	Demission	483mm×330mm×44.5mm (1U) 483mm×330mm×88mm (2U)
General	Temperature	0~45■(operation), -20~80■(storage)
General	Power Supply	AC100V±10%, AC 220V±10%, 50/60Hz Redundancy power supply (Optional)

Order Guide

Options	Configuration
Option 1	16 or 32 IP input, 16 or 32 analog channel outputs
Option 2	4 HDMI+28 IP input, 32 analog channel outputs
Option 3	8 HDMI+24 IP input, 32 analog channel outputs
Option 4	4 AV+28 IP input, 32 analog channel outputs
Option 5	8 AV+24 IP input, 32 analog channel outputs
Option 6	48 or 64 IP input, 48 or 64 analog channel outputs

System Construction Illustration



About Us

FMUSER is a company specializing in broadcasting and television transmission systems and integrating hotel IPTV systems. The company was established in 2009 and has served over 180 countries worldwide. More than 20,000 television stations, radio stations, and hotel clients from different.

Contact:Tomleequan Email■sales@fmuser.com whatsapp: +8613922702227

FMUSER 广州艾盛